

LUKOYANOV, Russian Federation

WMO: 276650

Lat: 55.033N

Lon: 44.500E

Elev: 718

StdP: 14.32

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-9.1	-19.8	1.9	-13.7	-14.0	2.6	-8.4	16.9	22.7	15.2	20.6	3.6	350	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	86.1	68.0	82.5	66.4	79.4	65.0	70.0	81.5	68.5	79.4	66.8	76.7	4.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.2	74.9	64.6	93.6	73.5	63.0	88.5	71.9	34.4	81.8	33.1	79.9	31.7	76.6	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.9	11.2	DB	-18.8	90.1	7.3	4.6	-24.1	93.4	-28.3	96.1	-32.4	98.7	-37.7	102.0	(4)
(5)			WB	-19.2	72.5	7.1	2.0	-24.3	73.9	-28.5	75.1	-32.5	76.2	-37.7	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	14.7	15.8	25.9	42.7	56.0	62.5	67.1	63.7	53.3	40.7	28.1	18.8	(6)
(7)		DBStd	21.03	12.30	12.20	8.44	9.54	9.17	7.73	6.69	7.59	7.87	8.15	9.68	11.82	(7)
(8)		HDD50	5087	1095	956	746	260	45	5	0	1	53	303	657	966	(8)
(9)		HDD65	9077	1560	1376	1211	670	302	137	56	116	359	753	1106	1431	(9)
(10)		CDD50	1776	0	0	0	41	232	380	529	426	152	16	0	0	(10)
(11)		CDD65	289	0	0	0	0	23	62	120	76	8	0	0	0	(11)
(12)		CDH74	2344	0	0	0	2	218	509	943	619	53	0	0	0	(12)
(13)		CDH80	629	0	0	0	0	36	127	262	199	5	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.6	5.7	5.5	5.3	5.0	4.4	4.0	4.0	4.2	5.2	5.5	5.7	(14)
(15)	Precipitation	PrecAvg	22.8	1.5	1.2	1.3	1.3	1.5	2.6	2.8	2.6	2.3	2.0	1.8	1.8	(15)
(16)		PrecMax	31.6	3.3	2.2	2.7	3.0	4.5	6.5	6.4	6.1	6.0	4.3	3.6	4.5	(16)
(17)		PrecMin	13.5	0.4	0.4	0.0	0.3	0.7	0.6	0.3	0.2	0.4	0.6	0.4	0.4	(17)
(18)		PrecStd	3.7	0.6	0.6	0.7	0.8	0.7	1.5	1.2	1.4	1.3	1.0	0.8	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	36.5	49.6	72.7	84.2	88.8	92.5	92.7	79.8	65.9	51.1	41.2	(19)
(20)			MCWB	33.9	33.9	42.5	54.4	63.8	66.4	67.8	68.5	61.5	54.4	47.8	39.9	(20)
(21)		2%	DB	33.5	34.6	43.1	67.5	79.7	83.5	85.9	85.4	74.5	59.7	45.7	37.0	(21)
(22)			MCWB	32.3	32.9	37.5	53.1	61.9	66.9	68.7	67.6	60.1	52.2	43.8	35.7	(22)
(23)		5%	DB	32.3	33.2	39.2	62.7	75.6	79.8	82.4	80.2	70.2	55.4	42.5	34.3	(23)
(24)			MCWB	31.3	31.9	35.3	50.6	60.1	65.2	67.4	65.7	58.8	50.0	40.8	33.1	(24)
(25)		10%	DB	30.8	31.9	36.5	58.2	71.5	76.2	79.3	76.3	65.9	52.0	39.5	32.6	(25)
(26)			MCWB	29.7	31.0	33.7	48.2	58.1	63.8	66.1	64.0	56.8	48.2	38.1	31.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.6	34.8	43.5	57.0	66.9	71.1	72.6	70.9	64.7	56.4	48.0	40.3	(27)
(28)			MCDWB	35.1	35.7	49.3	68.2	80.3	83.2	83.7	82.6	75.0	61.1	50.1	41.1	(28)
(29)		2%	WB	32.5	33.3	38.1	53.8	64.0	69.0	70.4	69.0	62.1	53.3	44.3	35.9	(29)
(30)			MCDWB	33.1	34.2	41.7	64.6	76.1	80.0	81.2	81.5	71.2	58.8	45.8	36.9	(30)
(31)		5%	WB	31.5	32.2	35.8	51.4	61.6	66.9	68.9	67.4	59.6	50.8	41.1	33.4	(31)
(32)			MCDWB	32.2	33.1	38.3	61.3	72.8	76.5	79.5	78.4	67.5	54.6	42.3	34.2	(32)
(33)		10%	WB	29.6	30.9	34.0	48.8	59.3	64.9	67.3	65.4	57.5	48.5	38.2	31.8	(33)
(34)			MCDWB	30.7	32.1	36.2	57.4	69.3	74.0	77.2	74.2	64.5	51.8	39.2	32.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	10.7	11.8	15.3	17.9	16.8	16.6	14.5	10.4	7.5	8.3	(35)	
(36)			MCDBR	7.3	8.6	13.8	21.5	22.4	21.1	20.6	21.7	20.9	16.7	9.4	6.3	(36)
(37)			MCWBR	7.4	7.9	9.5	11.3	10.2	9.2	8.3	8.9	10.0	10.3	8.5	6.3	(37)
(38)		5% WB	MCDWB	7.4	8.4	12.2	20.1	20.7	18.8	18.1	18.9	18.6	14.5	8.8	5.8	(38)
(39)			MCWBR	7.6	7.9	8.8	11.4	10.6	9.3	8.3	8.8	10.3	10.3	8.4	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.319	0.350	0.410	0.383	0.377	0.400	0.422	0.371	0.350	0.314	0.285	(40)	
(41)		taud	2.093	2.051	2.041	2.113	2.323	2.367	2.315	2.239	2.367	2.330	2.222	2.080	(41)	
(42)		Ebn at Noon	196	231	254	256	273	275	266	250	250	224	186	164	(42)	
(43)		Edh at Noon	21	31	40	43	37	36	37	38	29	24	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	231	526	1030	1308	1744	1807	1819	1446	897	429	171	126	(44)	
(45)		RadStd	37	74	119	125	122	167	170	147	127	67	32	35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.15	N/A	N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+1.06	N/A	N/A	N/A	N/A	N/A	-225	-311	N/A	N/A

Nomenclature: See separate page